



# Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI

# Dataset at a Glance

3,900

Purchases

18

Features

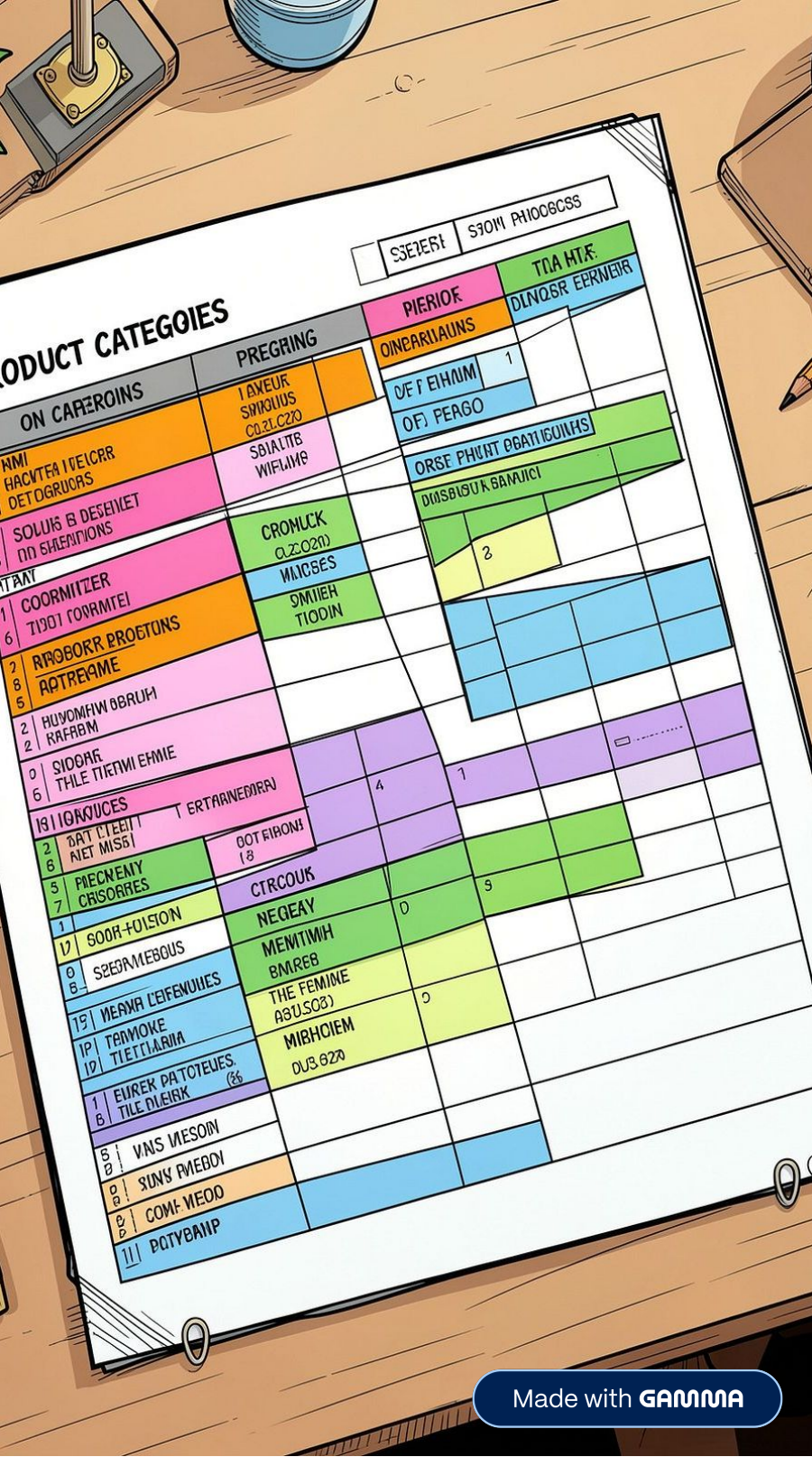
37

Missing Values

Review Rating only

\$59.76

Avg Purchase





# Data Preparation in Python



The image shows a person sitting at a desk, looking at a computer monitor. The monitor displays a code editor on the left and a data table on the right. The code editor shows Python code for data preparation, including column renaming and handling missing values. The data table has columns for various attributes and a 'Date' column. Red arrows point from the code to specific cells in the table.

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## → Clean & Standardize

Imputed missing Review Ratings;  
renamed columns to snake\_case

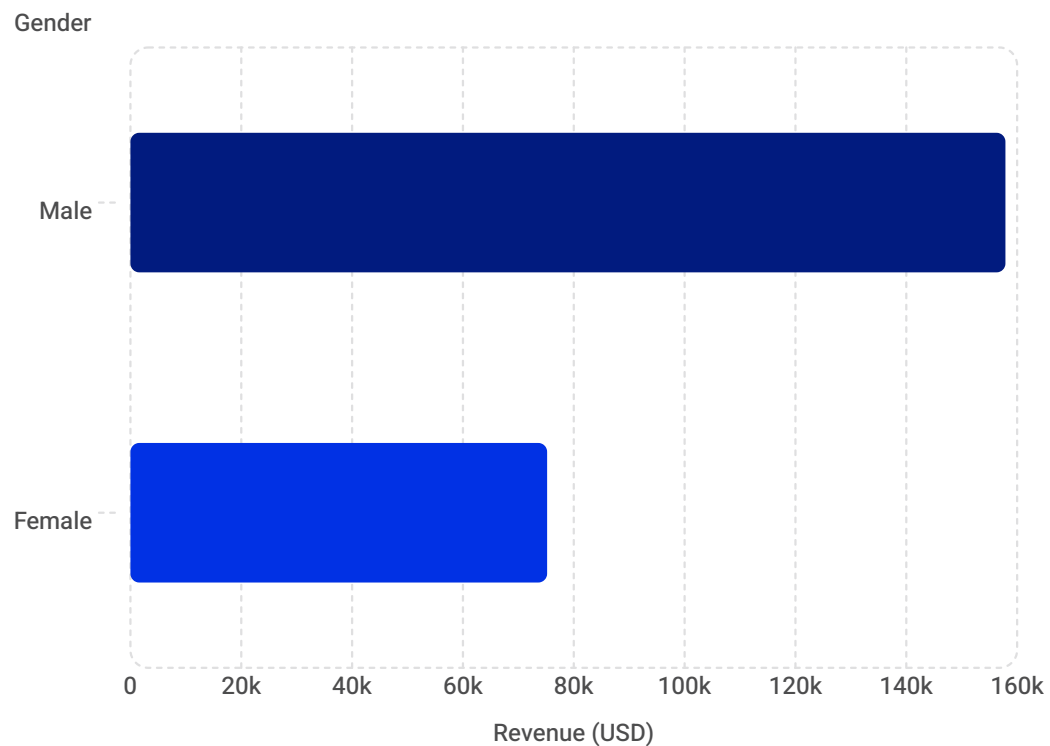
## → Feature Engineering

Created `age_group` and  
`purchase_frequency_days`

## → Load to PostgreSQL

Cleaned DataFrame loaded for SQL  
analysis

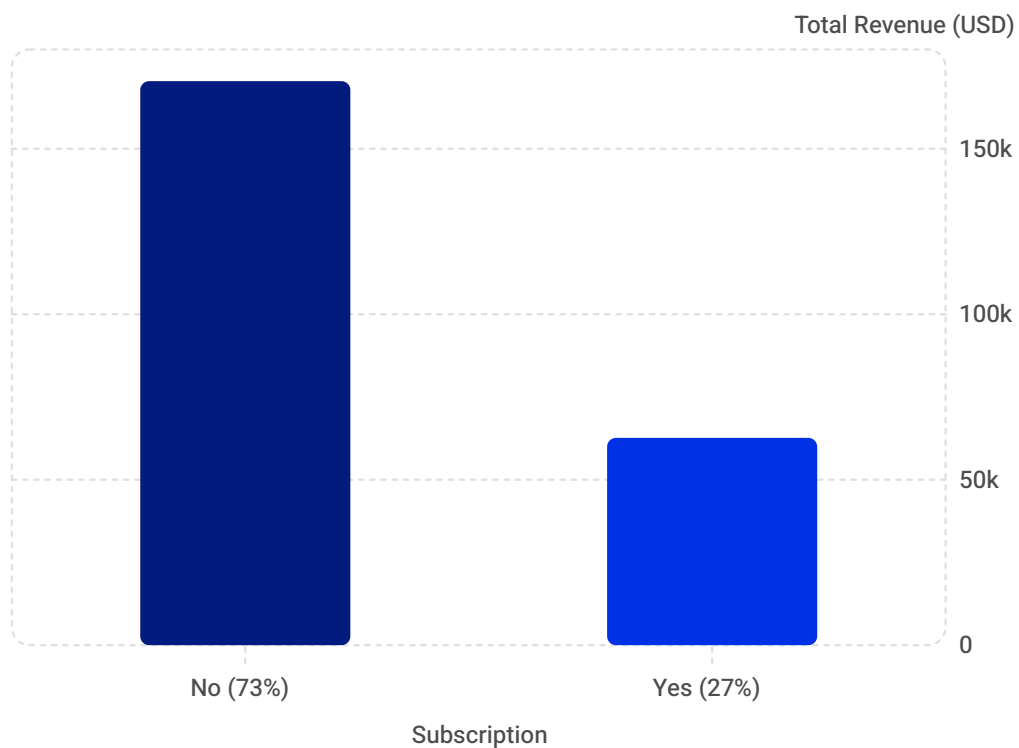
# Revenue by Gender



## Key Finding

Male customers generated **2.1× more revenue** than female customers across 3,900 transactions.

# Subscribers vs. Non-Subscribers



## Non-Subscribers

2,847 customers · \$170K  
revenue

## Subscribers

1,053 customers · \$63K  
revenue

❏ Similar avg spend (~\$59), but subscribers are only 27% of base



# Top Products by Category

1

## Clothing

Blouse · Pants · Shirt

2

## Footwear

Sandals · Shoes · Sneakers

3

## Outerwear

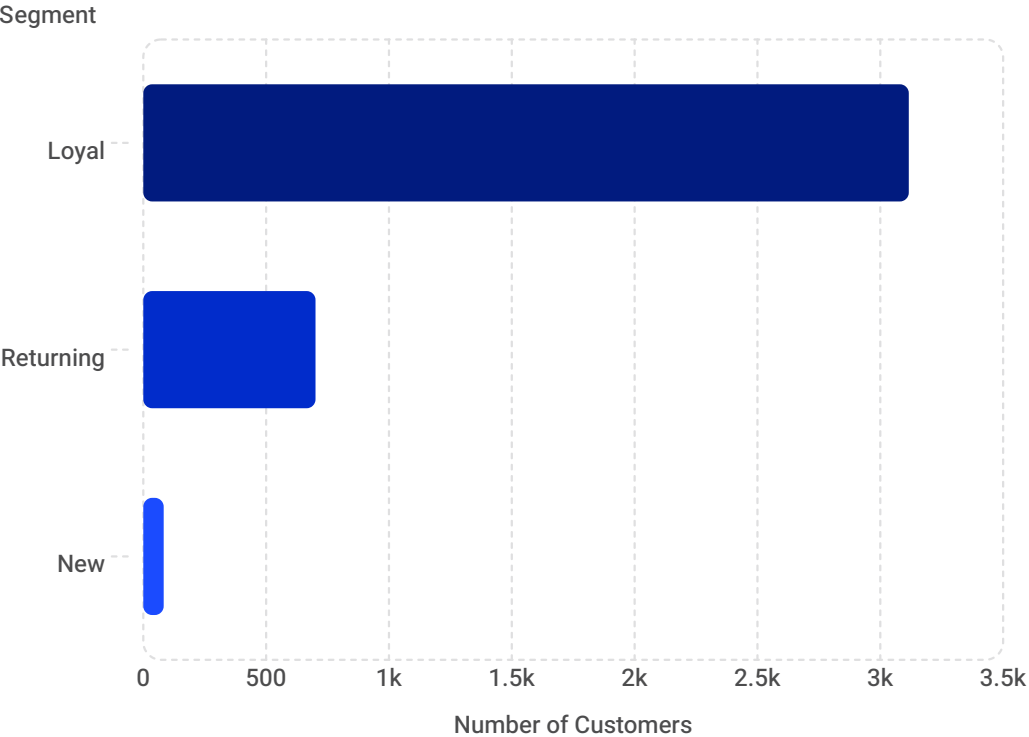
Jacket · Coat

4

## Accessories

Jewelry · Sunglasses · Belt

# Customer Segmentation



## Segment Breakdown

**Loyal — 3,116**

80% of customer base

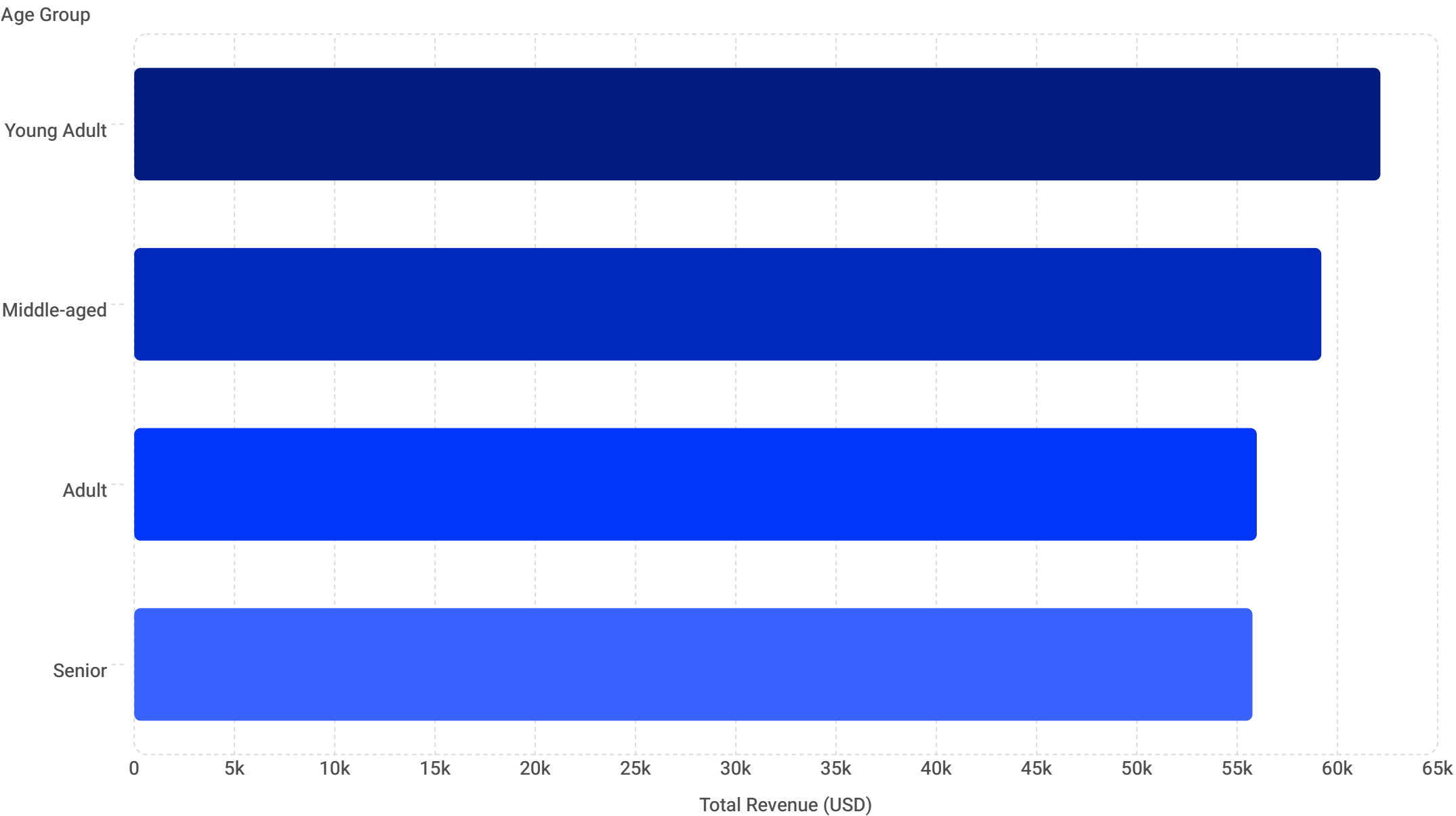
**Returning — 701**

18% of customer base

**New — 83**

2% of customer base

# Revenue by Age Group



Young Adults lead in revenue — a key target for marketing campaigns.





# Power BI Dashboard

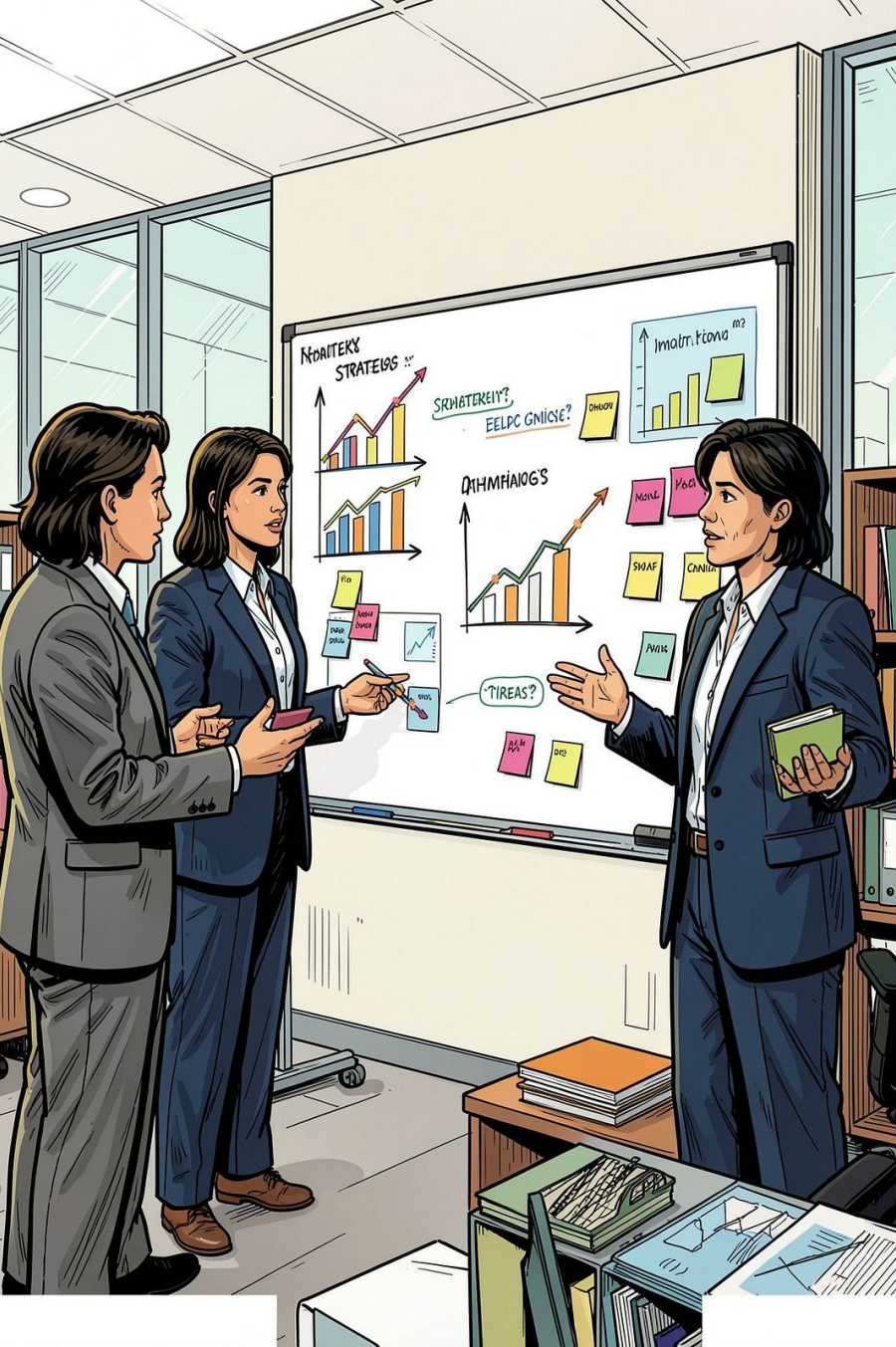
3.9K Customers

\$59.76 Avg Purchase

3.75 Avg Rating

Filters

Subscription · Gender · Category · Shipping



# Business Recommendations



## Boost Subscriptions

Promote exclusive subscriber benefits



## Loyalty Programs

Reward repeat buyers toward "Loyal" segment



## Review Discount Policy

Balance sales lift with margin control



## Targeted Marketing

Focus on Young Adults & express-shipping users